EXPERIMENT-02

AIM: Convert the E-R diagram into a relational Schema. Write complete MYSQL CREATE, TABLE statements with primary Keys, foreign Keys, NOT NULL, UNIQUE, ON DELETE CASCADE and SET NULL constraints. Insert Sample data demanstrate referential integrity violations.

Source Code:

```
DROP DATABASE IF EXISTS ecommerce_db;
CREATE DATABASE ecommerce_db;
USE ecommerce_db;

--1. CUSTOMER (Strong Entity)
 CREATE TABLE Customer1
        Customer_id INT AUTO_INCREMENT PRIMARY KEY;
        first_name VARCHAR(50) NOT NULL,
        last_name VARCHAR(50) NOT NULL,
        email VARCHAR(100) NOT NULL UNIQUE
    )
 ENGINE = InnoDB;
-- Multi-valued attribute: Customer phone numbers
 CREATE TABLE Customer_Phone (
        customer_id NOT NULL,
        phone VARCHAR(15) NOT NULL,
        PRIMARY KEY (customed_id, phone),
        FOREIGN KEY (customer_id) REFERENCES Customer
        (customerId)
            ON DELETE CASCADE
```

```
) ENGINE = InnoDB;

-- 2. ADDRESS (Weak entity - depends on Customer)
CREATE TABLE ADDRESS (
        Customer_id INT NOT NULL,
        address_id INT NOT NULL, -- partial Key
        street VARCHAR(100) NOT NULL,
        city VARCHAR(50) NOT NULL,
        State VARCHAR(50) NOT NULL,
        pincode VARCHAR(10) NOT NULL,
        PRIMARY KEY (customer_id, address_id), -- composite Key of weak entity
        customer(customer_id) ON DELETE CASCADE
) ENGINE = InnoDB;

-- 3. CATEGORY (Strong entity)
CREATE TABLE Category (
    Category-id INT AUTO_INCREEMENT PRIMARY KEY,
    category-name VARCHAR(50) NOT NULL UNIQUE
) ENGINE = InnoDB;

-- 4. SELLER (Strong entity)
CREATE TABLE Seller (
    Seller_id INT AUTO_INCREEMENT PRIMARY KEY,
    Seller_name VARCHAR(100) NOT NULL,
    email VARCHAR(100) NOT NULL UNIQUE.
) ENGINE = InnoDB;
```

```
CREATE TABLE Seller_Phone (
    Seller_id INT NOT NULL,
    phone VARCHAR(15) NOT NULL,
    PRIMARY KEY (seller_id, phone),
    FOREIGN KEY (seller_id) REFERENCES seller (seller_id)
        ON DELETE CASCADE

) ENGINE = Inno.DB;

--5. PRODUCT ( Strong entity with Specialisation)
CREATE TABLE Product (
    product_id INT AUTO_INCREEMENT PRIMARY KEY,
    product_name VARCHAR(150) NOT NULL,
    price DECIMAL(10,2) NOT NULL,
    Stock INT NOT NULL DEFAULT 0,
    category_id INT NULL,
    seller_id INT NOT NULL,
    FOREIGN KEY (category_id) REFERENCES category(category_id)
      ON DELETE CASCADE

) ENGINE = InnoDB;

-- Specialisation : Electronics (disjoint Subtype of Product)
CREATE TABLE Electronics (
    product_id INT PRIMARY KEY,
    warranty_period INT, -- in maths
    brand VARCHAR(50),
    FOREIGN KEY (product_id) REFERENCES Product
                                (product_id)
        ON DELETE CASCADE

) ENGINE = InnoDB;
```

```
-- Specialisation : Clothing ( disjoint Subtype of Product)
CREATE TABLE Clothing(
    product_id INT PRIMARY KEY,
    Size VARCHAR(10),
    fabric VARCHAR(50),
    FOREIGN KEY (product_id) REFERENCES product (productid
        ON DELETE CASADE

) ENGINE = InnoDB;

-- 6. ORDER (Strong entity) - "Order" is a reserved word in MySQL
CREATE TABLE orders(
        order_id  INT AUTO INCREEMENT PRIMARY KEY,
        total_amount DECIMAL(10,2) NOT NULL,
        order_date DATE NOT NULL,
        ~~total_amount~~ +
        customer_Id INT NOT NULL,
        FOREIGN KEY ( Customer_id ) REFERENCES Customer
        (Customer_id)

            ON DELETE CASCADE

)ENGINE = InnoDB

-- 7. ORDER_ITEM (weak entity - depends on olders)
CREATE TABLE Order Item(

            Order_id INT NOT NULL,
            item_no INT NOT NULL, -- partial Key
            quantity INT NOT NULL,
            unit_price DECIMAL(10,2) NOT NULL,
```

```
    PRIMARY KEY (order_id, item_no), -- composite key of weak entity
    FOREIGN KEY (product_id) REFERENCES orders (Order_id)
        ON DELETE CASCADE,
    FOREIGN KEY (product_id) REFERENCES product (product_id)
        ON DELETE SET NULL

) ENGINE = InnoDB;

-- 8. PAYMENT ( Strong entity, 1:1 with Orders)
CREATE TABLE Payment(
    payment_id INT AUTO_INCREEMENT PRIMARY KEY,
    amount DECIMAL (10,2) NOT NULL,
    payment_date NOT NULL,
    payment_mode (VARCHAR (20) NOT NULL,
    order_id INT NOT NULL UNIQUE -- unique enforces the
                                    1:1 Cardinality.
    FOREIGN KEY (Order_id) REFERENCES Orders (order_id)
        ON DELETE CASCADE

) ENGINE = Inno.DB,

-- 9. DELIVERY (Strong entity, 1:1 with Orders)
CREATE TABLE Delivery (
    delivery_id INT AUTO_INCREEMENT PRIMARY KEY,
    delivery_date DATE,
    Status VARCHAR (20) NOT NULL,
    tracking_number VARCHAR(30) UNIQUE,
    FOREIGN KEY (order_id) REFERENCES (order_id)
        ON DELETE CASCADE

) ENGINE = InnoDB,
```

```
-- SAMPLE DATA
INSERT INTO Customer (first_name, last_name, email) VALUES
    ('Meinal', 'Sharma', 'meinal.sharma@example.com'),
    ('priya', 'Iyer', 'priya.iyer@example.com');
INSERT INTO Customer_Phone (customer_id, phone) VALUES
    (1, '9436545210'),
    (2, '9123456700'),

INSET INTO ADDRESS (customer_id, address_id, Street, city,
                    State, pincode) VALUES

    (1, 1, '12MG Road', 'Lucknow', 'Uttar Pradesh', '226001'),
    (2, 1, '45 Anna Salai', 'Chennai', 'Tamil Nadu', '600002');

INSERT INTO (Category_name)
    ('Electronics'), ('Clothing'), '(Groceries');

INSERT INTO Seller (Seller-name, email) VALUES
    ('TechBazar Put Ltd', 'Contact@techbazar.com')
    ('fashionhub India', Support@fashionhub.in')

INSERT INTO PRODUCT (product_name, price, stock, category-
                    seller_id) VALUES

    ('wireless Earbuds X200', 2499.00, 50, 1, 1),
    ('Cotton Kurta - Blue', 899.00, 100, 2, 2);

INERT INTO Clothing (product_id, Size, fabric) VALUES
    (2, 'M', 'Cotton')

INSERT INTO ORDERS (ORder-Date, total_amount, customer-
VALUES ('2026-08-02', 3889.00, 1);
```

INSERT INTO ELECTRONICS (product_id, warranty_period, brand)
VALUES (1, 12, 'Soundwave')

INSERT INTO ORDER_ITEM (order_id, item_no, quantity, unit_price, product_id) VALUES

(1, 1, 1, 2499.00, 1),
(1, 2, 1, 899.00, 2),

INSERT INTO PAYMENT (amount, paymentDate, Payment_mode, order_id)

VALUES (3899.00, '~~Shipped~~' '2026-08-01', UPI, 1);

INSERT INTO Delivery (delivery_date, Status, tracking_number, Order_id) VALUES

('2026-08-05', 'Shipped', 'TRK1234567899', 1);

-- DEMONSTRATING REFERENTIAL INTEGRITY

(a) INSERT violation: Order_id 999 does not exists in Orders.

-- Expected : Error 1452 - Cannot add or update a child row.
a foreign key constraints fails (order_id. FK)

-- INSERT INTO Order_Item. (order_Id, item_no, quantity, unit_~~base~~ price)
product_id)

-- VALUES (999, 1, 2, 500.00, 1);

(b) INSERT VIOLATION : product_id 999 does not exist in product

-- EXPECTED : Error 1452 - foreign key constraints fail
(product_id, FK)

-- INSERT INTO Order_Item (Order_Id, Item_no., quantity, unit_price, product_id)

-- VALUES (1, 3, 1, 100.00, 999);

(c) UNIQUE Vailation : this order order already has a payment (1:1 constraint)

--EXPECTED : Error 1062_ Duplicate entry '1' for Key 'Order_id'

-- INSERT INTO Payment (amount, payment_date, payment_mode, order_id)

-- VALUES (500.00, '2026-08-02', 'Cash', 1);

(d) ON DELETE SET NULL demonstration : delete a Category

--EXPECTED : product. category_Id for its products become NULL,

-- DELETE FROM Category WHERE Category_id = 1;

-- SELECT * FROM product WHERE product_id = 1; -- category_id is now NULL.

(e) ON DELETE CASCADE demonstration : delete a Customer

--EXPECTED : their address, Customer_Phone, Orders and rows are deleted automatically, which is turn Cascade to Order_item, Payment and delivery for those Orders.

-- DELETE FROM Customers WHERE Customer_id = 1;

-- SELECT * FROM Order WHERE Customer_id = 1; -- empty result

-- SELECT * FROM Payment WHERE Order_id = 1, -- empty Result

(f) NOT NULL Vailation : Orders_date is required

--Expected : Error 1364 - field, 'Order_date' doesn't have a default value.

-- INSERT INTO Orders (total_amount, Customer_id) VALUES (500.00, 1);